

APPROVED METHOD:
SPECTRORADIOMETRIC
MEASUREMENT METHODS
FOR LIGHT SOURCES

AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-58-20 ERRATA

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Additions will be posted to this list online as they become available. This errata list was last updated **April 7, 2021**.

Additions are shown in **bold face**. Deletions are shown as ~~strikeout~~.

In first paragraph of Section 4.1.3 Bandpass:

The choice of bandpass is dictated by the purpose of the measurement. A narrow bandpass results in a low signal level, while a wide bandpass increases the error resulting from the spectral blurring associated with the bandpass. The action spectrum used to interpret the measured spectrum will determine the magnitude of error from spectral blurring due to a finite bandpass. The acceptable bandpass varies with the measurement's use. For example, in the UV photobiology, the S_{λ} action spectrum changes at a rate that requires a bandpass of 4 nm for a 2% error limit.[4] **[3]** This is considered a small error in safety evaluations (see ANSI/IES RP-27-20[2]), and therefore 4 nm or smaller is an acceptable bandpass for UV photobiology measurements. For illumination measurements, photometry, and colorimetry, the bandpass of an instrument shall not exceed 5 nm in order to limit the measurement errors resulting from the finite bandpass when measuring a variety of sources, including discharge lamps.[3] In the case where the bandpass exceeds 5 nm, the errors in the measured spectra resulting from the large bandpass should be corrected before calculating photometric and colorimetric values using the method described in CIE 214:2014.[15]**[14]**

In last paragraph of Section 6.2 Bandpass Correction:

A third method is the extended method for bandpass correction, published by Ohno[11] and Gardner.[12] In real instruments, the instrument bandpass is not a triangular shape, and its width does not match the scanning interval. It is often seen that the bandpass of a spectroradiometer, designed to be constant, can vary as much as 20%. At 20% deviation, the reduction of the error with these methods will be about half or less. In some cases, the bandpass function might resemble a Gaussian function, a trapezoid, or an asymmetric triangle, which would also add deviation from the requirement. In such conditions where the bandpass and scanning interval are not well matched, ASTM E308[10]**[9]** or the Stearns and Stearns method[11]**[10]** cannot be used. This improved method is now available to be applied to any arbitrary bandpass function. As this is a numerical approach, the solution can be given for any shape of bandpass, which may be nonlinear, asymmetric, and/or not matched with the scanning interval, with the assumption that the measured spectrum is a smooth function (quadratic within the bandpass). This method also provides accurate wavelength calibration for an asymmetric bandpass (often found in array instruments).

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Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared by
The IES Testing Procedures Committee



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Foreword

This guide is a revision of *IES LM-58-15, IES Guide to Spectroradiometric Measurements* with editorial changes only.

1.0 Introduction and Scope

1.1 Introduction

In evaluating the color performance of light sources, there are two factors to be considered:

- The light source color appearance (hue and saturation)
- Its color rendering ability (the effect of the light source on the appearance of objects, compared to their appearance under a reference source)

The perceived color of a light source is its appearance as a separate consideration from its luminance, geometry, and time variations. This in turn depends upon the relative spectral power distribution (SPD) of the radiant energy from the source and on observer adaptation. Two sources with entirely different spectral power distributions may produce the same color appearance when illuminating one surface (i.e., they produce a metameric match) but can look quite different when illuminating a surface with different spectral reflectance characteristics.

The SPD constitutes the basic data needed for the computation of chromaticity coordinates and the metrics that describe color rendering. These data are obtained by spectroradiometry, whereby the light from the source is dispersed into its component wavelengths and the power in each narrow band of wavelengths is measured. Spectroradiometry has been practiced since Newton discovered the dispersive properties of a prism on light. Spectroradiometry is now recognized as the most accurate, precise, and dependable method of determining the chromaticity of any light source.

Where visual phenomena such as color are involved, the SPD usually is determined between the wavelengths

of 380 nm and 780 nm, and should be measured with sufficiently narrow bandpass to show the desired resolution. For more-complete coverage, the SPD should be determined between the wavelengths of 360 nm and 830 nm. The curves are often plotted at five-nanometer intervals and are based on a spectral bandpass of approximately five nanometers. Computer plotted spectral power distributions are common at intervals much smaller than five nanometers. For sources containing line spectra, it has been found necessary for accurate resolution of spectra to utilize a bandpass of approximately two nanometers.

All photometric, colorimetric, and radiometric values can be calculated using the SPDs measured by a spectroradiometer. Sufficient accuracy is attainable with modern spectroradiometric systems such that they can be used for defining standards for other methods of colorimetry (e.g., with tristimulus colorimeters). In general, a spectroradiometer consists of a monochromator (dispersing instrument), a detector to measure the power at the output of the monochromator, and a device for recording data. A standard light source of known spectral power distribution is used for calibration.

The spectroradiometric method is advantageous because once calibration has been carried out against one standard source (usually an incandescent filament lamp), the spectroradiometer can determine—with equal accuracy—the SPD of light sources of any color, providing only that the unknown SPD is within the range of calibration. This range can comfortably run from 380 nm to 780 nm—or preferably from 360 nm to 830 nm—and therefore covers the visible region of interest for determining chromaticity.

1.1.1 Safety Precautions. Since overexposure to bactericidal (and/or germicidal) and erythematous ultraviolet radiation may result in erythema (reddening of the skin) or keratoconjunctivitis (inflammation of the cornea and conjunctiva, the exposed eye membranes), it is imperative that suitable eye protection and clothing be used when conducting measurements in these spectral regions. Lamp manufacturers' recommendations for safety precautions should be followed.¹

Strong ultraviolet sources, such as deuterium lamps, can produce ozone. Adequate ventilation should be maintained.

1.2 Scope

This document describes the requirements and recommendations of the instruments and procedures for spectroradiometric measurements, including those of color performance, spectral irradiance, spectral radiance, and spectral total radiant flux, either in relative or absolute units. The spectral range is from approximately 200 nm to 1,700 nm, where the characterization of light from light sources, visual displays, and light emitting diodes is most commonly done. This document does not provide in-depth detail on every subject but directs the user to references that completely describe the concepts. The light source or device under test shall be operated in accordance with the appropriate IES Light Measurement (LM) or American National Standards Institute (ANSI) document pertaining to the device; such operation is not described in this document.

2.0 Normative References

There are no normative references.

3.0 Definitions and Nomenclature

colorimeter: An instrument designed for measurement of the chromaticity of a light source without determining its SPD. It may give chromaticity coordinates directly or, by suitable calibration, give data that can be converted to these coordinates.

input optics of a monochromator: The optical elements placed between a light source and the entrance slit of the monochromator to direct radiation of a light source to the monochromator.

slit scattering function (bandpass function): The relative spectral throughput function of a monochromator or relative spectral responsivity function of a spectroradiometer at a fixed wavelength

setting with respect to the wavelength of the incident monochromatic source.

total spectral radiant flux, $W \cdot nm^{-1}$: The geometrically total radiant flux in all directions (4π steradians, sr) from a source, at a given wavelength, per unit wavelength.

4.0 Spectroradiometers

4.1 Characteristics

4.1.1 Wavelength Dispersion. The monochromator may be either a single-pass or double-pass instrument. The dispersing element(s) may be a prism, a diffraction grating, or a combination of the two. Prisms are usually less expensive, but their dispersion varies with wavelength such that longer wavelengths are crowded closer together in the monochromator output. A uniform wavelength interval (bandpass) is desirable for easy reduction of the data. Diffraction gratings have uniform dispersion but give rise to higher orders of diffractions. Higher orders shall be eliminated. For example, use of a filter eliminating all wavelengths shorter than 370 nm yields a “clean” spectrum from 370 nm to 740 nm, and this range serves most purposes for colorimetry of light sources.

4.1.2 Wavelength Calibration. It is recommended that the expanded uncertainty of the wavelength calibration should be ± 0.1 nm ($k=2$). The expanded uncertainty of the wavelength scale shall be less than ± 0.5 nm ($k=2$). The wavelength scale should be calibrated periodically. To satisfy these requirements, it is usually necessary that the ambient temperature be controlled within ± 1 °C. The wavelength scale can be calibrated by measuring the emission lines of known wavelengths from low-pressure mercury lamps (253.7 nm, 365.0 nm, 404.7 nm, 435.8 nm, 546.1 nm) and neon-discharge lamps (614.3 nm, 650.6 nm, 703.2 nm). For grating type or prism-grating type monochromators, the wavelength scale can be checked at a few points since the scale is normally linear. For prism monochromators, the scale should be checked at several points and interpolated over the entire wavelength region since the wavelength scale is nonlinear.

4.1.3 Bandpass. The choice of bandpass is dictated by the purpose of the measurement. A narrow bandpass results in a low signal level, while a wide bandpass increases the error resulting from the spectral blurring associated with the bandpass. The action spectrum used to interpret the measured spectrum will determine the magnitude of error from spectral blurring due to a finite bandpass. The acceptable bandpass varies with the measurement's use. For example, in the UV photobiology, the S_A action spectrum changes at a rate that requires a bandpass of 4 nm for a 2% error limit.⁴ This is considered a small error in safety evaluations (see ANSI/IES RP-27-20²³), and therefore 4 nm or smaller is an acceptable bandpass for UV photobiology measurements. For illumination measurements, photometry, and colorimetry, the bandpass of an instrument shall not exceed 5 nm in order to limit the measurement errors resulting from the finite bandpass when measuring a variety of sources, including discharge lamps.³ In the case where the bandpass exceeds 5 nm, the errors in the measured spectra resulting from the large bandpass should be corrected before calculating photometric and colorimetric values using the method described in CIE 214:2014.¹⁵

Bandpass is typically quantified using the spectral distance between the two 50% points of the slit scattering function (SSF; see **Section 4.1.4**), known as full-width half-maximum (FWHM). This scalar quantity is a simplification that does not consider the shape. SSF shapes vary from rectangular polychromators (array systems) with slit image larger than the pixel size, to shapes that are smoothed and asymmetric from the systems that have a poor quality of slit image due to spherical and/or coma aberrations.

4.1.4 Slit Scattering Function (Bandpass Function).

The slit scattering function (SSF) of a monochromator is determined by both the entrance slit width and the exit slit width. The SSF around the bandpass is normally symmetrical and trapezoidal in shape, centered around the wavelength setting and falling to zero on both sides. For measurements of discharge lamps, the SSF should be symmetric and preferably triangular in shape. If the SSF is not symmetrical, the wavelengths measured by emission lines may not agree with the effective wavelengths for continuous spectra. The SSF will be made closest to a triangle by adjusting the entrance-

slit bandpass and the exit slit bandpass to be the same value. In principle, measuring an SSF requires a tunable laser. In practice, however, an SSF may be obtained from its corresponding line spread function (LSF). Within the bandpass, an SSF may be approximated as the mirrored function of its corresponding LSF. An LSF is a relative output signal function of a monochromator when measuring a monochromatic light source, which can be obtained by measuring a line spectrum from a laser or a low-pressure mercury lamp, and by scanning the monochromator at very small wavelength intervals, such as 0.1 nm, around the emission line wavelength.

4.1.5 Wavelength Interval. The wavelength interval is the increment in wavelength at which the detector output is sampled for a scanning spectroradiometer. The response of the system per wavelength is the superposition of the SSFs at the wavelength intervals. In a perfect system with a triangular SSF and constant bandpass, the wavelength interval should be set equal to the bandpass. For a system with a triangular SSF but bandpass varies across its spectral range, the wavelength interval should vary so it matches with the bandpass at any wavelength.

When the SSF that is not triangular, the scanning interval should be reduced to an integer fraction of the bandpass (e.g., 1/5, 1/3, or 1/2) to reduce the errors in calculated photometric and colorimetric values; for example, for a wavelength interval of 1 nm and a bandpass of 5 nm, the error in measured photometric values resulting from the non-triangular SSF will be limited to 1% in most cases. Alternatively, one can set the wavelength interval equal to the bandpass and correct the errors in the measured spectra resulting from the non-triangular SSF before calculating photometric and colorimetric values using the method described in CIE 214:2014.¹⁵

4.1.6 Stray Light. The stray light of a monochromator is the radiation at wavelengths outside of the selected bandpass that pass through the exit slit of the monochromator. Manufacturers of monochromators usually specify the level of stray light. Well-designed monochromators will have stray light less than or equal to 0.0001 times the maximum signal for each monochromator stage or pass. Thus, the stray light for a good double-pass monochromator is less than 10^{-8} times the maximum signal.

To evaluate stray light, radiation of a line spectrum of wavelength λ_0 with no continuous spectral components, such as from a helium-neon (He-Ne) laser beam, is directed to the input optics and measured as a test source for the entire spectral region of the monochromator. (Note that when measuring stray light, the noise level shall be lower than the signal to be measured.)

In general applications, it is desirable that the stray light should be less than 0.0001 times the source signal, which would correspond to errors less than 0.001 in (x, y) chromaticity coordinates in fluorescent lamp measurements. Care should be taken to reduce stray light, especially with single-pass monochromators. Filtering out radiation that is outside the wavelength region to be measured is usually a good method for reducing stray light (e.g., infrared radiation of incandescent lamps can be filtered out for measurements in the visible region). In addition, in grating monochromators, higher order spectra (second order is dominant) shall be rejected by using long-pass filters or another dispersive element. For example, a long-pass filter cutting off at 350 nm should be used for measurement of 380 nm to 700 nm, and a long-pass filter cutting off at 600 nm should be used for measurements of 700 nm to 1,200 nm. Prism or prism-grating monochromators do not need such manipulations.

When calibrating a monochromator-detector pair with a halogen or incandescent source, the stray light component may be significant in the blue-UV range of the calibration curve. It should also be noted that stray-light error can be significant for measurements where the spectral power of the source is very small or even zero.

For accurate spectral measurements in the UV, double-pass monochromators are usually required. Double-pass monochromators give a stray light error on the order of 10^{-8} , which is necessary because the source signal in the UV-B region is often less than 0.0001 times that in the visible region. This is particularly true for an incandescent source, usually used to calibrate the spectrometer.

4.1.7 Input Geometry. From the definitions of spectral irradiance, spectral radiance, and spectral total radiant

flux,⁴ it follows that light from a source or an object surface should be guided to the monochromator or spectroradiometer such that the light can be measured in the units that correspond to these metrics. Measurements are normally performed as the comparison of a test lamp with a standard lamp having a known spectral power distribution. It is essential that the effective receiving area of a dispersive element (such as a grating) be completely and uniformly irradiated no matter the size or light distribution of the source being measured. It is also necessary to depolarize the input light to eliminate polarization errors. The input optics shall benefit the source measurement application.

A fiberoptic bundle coupled with the spectroradiometer or monochromator allows a flexible method to couple the measurement target area to the spectroradiometer.

4.1.8 Calibration Sources. The calibration sources used, herein referred to as *standard sources*, shall be traceable to the International System of Units (SI) via a national metrology institute (NMI). Calibration sources are generally created from deuterium lamps for the UV region, and from incandescent or halogen lamps for the UV-VIS-NIR (visible and adjacent ultraviolet and near-infrared) regions. Recalibration at regular intervals is essential to maintain traceability. It is preferable to have at least three lamp standards of any one type to permit cross checks to detect drift in the lamp characteristics. The use of a standard source whose characteristics are similar to those of the test source is advantageous in that some systematic errors tend to cancel out because of like-to-like comparisons.

4.1.8.1 Spectral Irradiance. For the visible and infrared regions, the standard sources shall be incandescent lamps (including tungsten-halogen lamps) or integrating-sphere sources using incandescent lamps. Standard lamps, calibrated in $W/(m^2 \cdot nm)$, should be mounted in special bases, such as a medium bi-post base, to improve alignment accuracy, which minimizes uncertainties due to positioning and orientation of the standard lamp. For the UV region, tungsten-halogen lamps (above 250 nm) and or deuterium lamps (200 to 400 nm) are recommended. The signal output of deuterium lamps is not as stable as that of incandescent lamps. Thus, the uncertainty of the absolute irradiance of a standard deuterium lamp can be reduced by a factor

of approximately two by scaling the spectral irradiance of the deuterium lamp to the spectral irradiance of a tungsten-halogen lamp standard over the wavelength range 250 nm to 300 nm each time the lamp is operated.

For spectral irradiance measurements, the radiation of a test source and a standard source shall be directed to one of the following optical systems:

- An integrating sphere; the diffuse radiance of the sphere wall is collected at the entrance slit of the monochromator
- A light-reflecting diffuser made of such material as barium sulfate or pressed or sintered polytetrafluoroethylene (PTFE) irradiated normally and viewed at 45 degrees
- A light-transmitting diffuser such as a ground quartz plate; the diffused transmitted radiation is directed to the entrance slit of the monochromator

4.1.8.2 Spectral Radiance. The standard sources shall be tungsten-ribbon filament lamps, integrating sphere sources, blackbodies, or light diffusers whose spectral radiance factors are known and which are irradiated by a calibrated spectral irradiance standard lamp (see **Section 4.1.8**). Each type of standard source should have a mark that permits reproducible angular positioning.

For spectral radiance measurements, the input optics shall be composed of an imaging system that directs radiation only from a limited solid angle (measurement angle). Stray light in the imaging system should be low enough so that radiation from outside the measurement angle will not affect the measurement results. Spectral radiance sources are calibrated in $W/(m^2 \cdot sr \cdot nm)$.

4.1.8.3 Total Spectral Radiant Flux. The standard sources shall be incandescent lamps, including tungsten-halogen lamps. For total spectral radiant flux measurements, the standard source, calibrated for W/nm , and the test source are alternately placed and operated in an integrating sphere designed for lamp measurement. The uniform radiation of the sphere exit port is introduced to the monochromator entrance slit. For array systems, an optical fiber with a cosine correcting material positioned on the sphere wall surface is typically used. (For more information on sphere measurements, refer to ANSI/IES LM-78-20.⁵)

An alternative method of calibrating a sphere system for total spectral radiant flux is to use a spectral irradiance standard and a total luminous flux standard. With the test lamp mounted in the sphere, the light from the spectral irradiance standard is allowed to enter the sphere from a side port where the created beam of light contacts the sphere wall surface only. The detector shall be baffled so as not to receive light from this "hot spot." The relative spectral responsivity of the sphere system is then calibrated using this source, including the self-absorption effects of the test lamp. The relative spectral responsivity calibration is then corrected to an absolute responsivity by using a total luminous flux standard.

For more information on the operation and electrical measurement of light sources, the reader is referred to the numerous lamp measurement (LM) guides available from the IES.

4.2 Array Spectrometry

An array spectrometer can carry out measurements much faster than a scanning spectrometer because the entire wavelength range is recorded simultaneously rather than sequentially. The fastest measuring times are around 6 milliseconds (ms), and an increase in sensitivity can be achieved by increasing the integration (measurement) time. The largest drawback of array spectrometers is that they are made as single-pass monochromators. As a result, spectral stray light level is high and optical dynamic range is limited to 3 or 4 orders of magnitude, where the dynamic range achieved by a mechanically scanning spectroradiometer employing a double-pass monochromator configuration is twice as large. The spectral resolution of array spectrometers is in the range from 1 nm or less to 10 nm or larger, depending on the number of pixels of the detector and the optical resolution of the spectrograph.

The imaging optics in the spectrometer are extremely important to guarantee optimum flat-field correction for the flat detector. Therefore, optical imaging errors such as astigmatism and field curvature shall not cause any image distortions over the entire width of the detector. Imaging errors of this type lead to deterioration of the spectral resolution at the outer edge of the detector. Improved imaging characteristics can generally be obtained with longer focal lengths and larger optics.

4.2.1 Design. Charge-coupled device (CCD) arrays show a higher sensitivity, but diode arrays offer a better dynamic range. Proper selection of the detector array for the application can improve the performance of the whole system, or can just make it possible. For near-monochromatic light-emitting diode (LED) sources, with the total output enveloped under a near-Gaussian spectral distribution, CCD arrays are suitable (see **Section 4.2.1.1**). For light measurement of sources with non-continuous spectra, such as fluorescent and HID sources, where the total output is distributed over the visible but the strong peaks dictate the dynamic range of the sensors, the diode array is more suitable.

4.2.1.1 CCD Array Spectrometers. A CCD spectrometer uses a charge-coupled device to measure the energy of incident photons. CCD spectrometers can be sensitive to a wide range of wavelengths, including infrared, optical, ultraviolet, and X-rays. Each incident photon generates electrons in the CCD, the number of which is proportional to the energy of the original photon. The robust construction of array spectrometers and their short measuring time make them ideal for production applications. The CCD sensors used in array spectrometers can range from photocopy grade to thermal-electrically cooled high-quality "back-thinned" sensors. Array spectroradiometers with this type of detector are now preferable for production control of LEDs because they eliminate the disadvantages of photometers without compromising on measurement time. The advantages for the CCD array spectrometers are many pixels (2,048 in the wavelength direction), high sensitivity, high speed, higher response in the blue, and better sensitivity than a photodiode array.

4.2.1.2 Linear Diode Array Spectrometers. A silicon photodiode array consists of a linear array of multiple photodiode elements. Each pixel consists of a P/N junction with a positively doped *P* region and a negatively doped *N* region. When light enters the photodiode, electrons become excited and generate output as an electrical signal. Most photodiode arrays have an integrated signal processing circuit with an integration and readout amplifier on the same chip. The advantages for the photodiode array are high near-infrared sensitivity, high speed, low noise, and better baseline stability than CCDs for longer integration times.

4.2.2 Integration Time. Integration time is the period during which the spectrometer's detector is exposed to light. An integration time of 1 ms is the typical lower limit for most CCD based spectrometers, while 10 microseconds (μ s) is the typical lower limit for on-chip electronic shuttering. Shorter integration time means faster online device test processing as well as greater dynamic range for measuring sources of high luminous intensity.

The integration time shall be chosen to ensure that the detector does not become saturated. In addition, the integration time shall be long enough to ensure that for devices not having a constant optical output (i.e., for those with flicker), several cycles are sampled.

4.2.3 A/D Converter Data Rate. The A/D converter data rate represents how fast the data can be read out of a CCD spectrometer and into the computer for processing. This value, along with the number of pixels in the array and the integration time lead to how many scans/second can be achieved. This is important for on line testing.

4.2.4 Dynamic Range. The dynamic range of a system is defined as the full-scale signal divided by the minimum resolvable signal. For spectrometers, the minimum resolvable signal can be defined as the standard deviation of the dark signal.

A common misunderstanding is to confuse the dynamic range with the A/D converter's resolution. However, a dynamic range measurement includes the system's minimum noise level (i.e., detector readout noise, electronic noise, and quality of the design).

Dynamic range is most applicable to low light level applications, like fluorescence, because it defines the minimum detectable signal. It is important to realize that a spectrometer's dynamic range is a measurement of its performance that does not include light (i.e., not the way it is used). Thus, the signal-to-noise ratio is typically a more meaningful metric (see **Section 4.2.5**).

4.2.5 Signal-to-Noise Ratio (S/N Ratio). The S/N ratio measures how precisely the spectrometer can measure a certain light level. The S/N ratio is the average signal level

divided by the standard deviation of the measured signal at a certain light level. While S/N ratio varies with the light level, most manufacturers specify it at its maximum level, which is the full-scale illumination. Ideally, the spectrometer should only be limited by the optical shot noise, which indicates that S/N ratio increases as the square root of the number of photons collected.

4.2.6 Linearity. Two types of linearity are present in CCD and diode-array spectroradiometers. The first type is defined as the efficiency of the detection device in turning photons into captured charge. As the detector well fills with charge, the efficiency of the photon-to-charge conversion changes. Ideally, any change in the light flux launched into the spectrometer leads to a proportional change in the detector signal; otherwise, the system is not suitable for radiometric measurements. CCD-based spectrographs are inherently non-linear. It is recommended to use the manufacturer's non-linearity correction coefficients if available.

The second type of linearity is associated with the signal with respect to the integration time. The uncertainty associated with the integration time will be larger for short integration times than for longer ones. The effect is referred to as "smearing." The best way to avoid smearing effects is to use integration times similar to the calibration integration time.

4.2.7 Dark Measurements. Dark scans shall be subtracted from the light scans to get a signal value that is representative of the light signal being measured. Dark noise should not be confused with baseline noise; it is the accumulation of charge in the pixel wells due to thermal effects at the quantum level. This value is dependent on both the temperature and the integration time. To reduce temperature-induced charge, thermal-electric coolers are often used to increase the usable integration time. It is important to note that the dark signal will eventually reach saturation level if the integration time is long enough.

4.2.8 Spectral Resolution. In general, this figure of merit is a function of the wavelength range and slit width. If two spectrometers have identical wavelength ranges (grating groove density) and slit widths, then better resolution is related to the quality of the optical

system. This value can be wavelength dependent and should be measured over as much of the usable range as possible. This is done by measuring the full-width half-maximum (FWHM) of various spectral peaks of a line source, such as a mercury (Hg) or argon (Ar) lamp.

4.2.9 Trigger Capability. Triggering is a feature that includes a set of I/O functions that allow an array spectrometer to coordinate its scan time with the output of the power supply for the device under test. Array spectrometers can be triggered through software or hardware.

4.2.10 Wavelength Uncertainty. Wavelength uncertainty is governed by two things: the dispersion linearity of the grating, and the quality of the wavelength calibration. The more linear the grating dispersion, the easier it is to fit a polynomial curve to match pixel number with wavelength. The quality of the wavelength calibration is determined by a careful selection of spectral lines used in line sources such as helium (He), Hg, and Ar. Care shall be taken to avoid lines that are too close to each other, which will skew the apparent peak wavelength. In addition, inter-pixel interpolation of the peak position shall be used. Wavelength measurements should be based on centroid of measurements.

4.2.11 Bandpass Determination. Because the entrance slit and detectors in an array spectrometer are fixed, the measurement of the bandpass requires the use of a narrowband tunable laser. Instead of setting the entrance and exit slits and scanning the monochromator, as is done in the determination of the slit scattering function (refer to **Section 4.1.4**), the narrowband tunable laser is scanned and the signal is collected from one pixel.

An alternative method is to measure the spectrum of a single narrowband emission line or a narrowband laser line, and invert the measured spectrum. When a narrowband line is measured, the spectrum is equivalent to the reverse bandpass function. If the pixel count is large (wavelength interval is small), the measured spectrum can be inverted about the center frequency of the emission line. If the pixel count is sparse (wavelength interval is large), the inverted spectrum can be fit to a model function that is described in Ohno et al.⁶

4.3 Scanning Spectrometry

With mechanical scanning spectroradiometers, the standard source is first measured throughout the entire wavelength region and then test sources are measured. In this method, it is recommended that the standard source be measured both before and after the test sources are measured, unless the historic stability of the spectroradiometric system indicates otherwise. During the entire measurement period, care should be taken to maintain all of the measurement conditions as constant as possible, and one period should not exceed several hours. By checking the difference between the data of standard sources "before" and "after," the stability of the spectroradiometer system during the measurement period can be validated. The average of two scans for the standard source may be used to determine the spectral calibration factors.

Array spectroradiometers are generally more stable over time than scanning spectroradiometers and will not require calibration each time a measurement is made. However, array spectroradiometers need to be calibrated with a standard source periodically, and frequently enough that the drifts of responsivity over time are within the required uncertainty.

With mechanical scanning spectroradiometers, when the dark signal of the detector is unstable while the responsivity stays constant, the light that is entering the monochromator should be chopped, and a lock-in amplifier (AC amplification) should be used. In this case, the response of the detector and its amplifier circuit shall be fast enough for the chopping frequency. The possibility of amplification rollover can be checked by measuring the data at several different chopping frequencies. The chopping frequency should not be a multiple of the AC line frequency. The use of a chopper with AC discharge sources is not recommended since beat modulation may corrupt test data.

When test sources are not reproducible, it is desirable to measure the same test sources three times or more, and to report the mean value and the standard deviation for spectral quantities and/or chromaticity coordinates.

4.3.1 Continuous Scanning. Some mechanical scanning spectroradiometers are designed so that measurements

are made while the wavelength is being scanned. In the continuous scan, the scan speed shall be slow enough and the data sampling time short enough such that the effective bandpass is not affected. The data sampling timing should be adjusted to compensate for the delay caused by the wavelength scan during the sampling time.

The time constant of the current-to-voltage converters and other amplifiers shall be fast enough to respond to the sharp change of detector signal at each wavelength scan, especially with continuous wavelength scanning.

For fluorescent lamps, a scanning method has been used in the past in which the power of emission lines is measured separately from the phosphor feature. This method, however, is not recommended because many phosphors used in modern fluorescent lamps are also narrowband emitters, and in many cases, it is almost impossible to separate the line spectrum of the Hg from that of the phosphor, e.g., the 546-nm Hg line and the green phosphor.⁷

4.3.2 Sequential Scanning. When sequentially scanning, the time delay between wavelength steps shall be long enough for the detector signal to settle and be recorded. This is especially important when performing low light level measurements employing a lock-in amplifier with large sample averaging.

When both the standard source and the test source are very stable over a long period, and when highest accuracy is desired, the two sources can be turned on at the same time and measured in turn at each wavelength while the monochromator pauses at each wavelength. In this method, any drift of spectroradiometer parameters, including drift and fatigue of the detector, will be canceled out. However, an automatic means to switch the monochromator input between two sources is necessary.

The time constant of the current-to-voltage converters and other amplifiers shall be fast enough to respond to the sharp change of detector signal at each wavelength

Depending on the applications, the bandpass and the scanning interval need to be matched for sequentially scanned spectra; for example, for measurement of discharge lamps having emission lines, or to use

bandpass error correction techniques. Thus, the data interval is often tied to an instrument's bandpass and cannot be discussed separately. (Refer to Chapter 5 in *Colorimetry: Understanding the CIE System* for a complete discussion and analysis of this topic.⁸⁾

5.0 Detectors

5.1 General

In this section, several detectors are described that can be coupled to a monochromator. This is not an exhaustive list but includes many of the popular detectors used to cover the range of 200 nm to 1,700 nm. When using any type of detector, compensation shall be made for the dark signal, including that of the amplifier, either by measurement and subtraction from the signal output or by electronically adjusting the zero point of the detector readout system.

In order to avoid errors due to nonlinearity of the detector, it is desirable to check the linearity. It is also desirable to set the signal levels for the test source and for the standard source in the same range, to minimize the effect of nonlinearity of the detector.

For a photovoltaic detector, the spectral response typically changes as the temperature changes. As the temperature decreases, the overall sensitivity increases. For photoconductive detectors, the spectral response, dark resistance, and time constant change with the ambient temperature. To reduce the effects of ambient temperature, measurements in the near infrared should be performed with a cooled detector. Many detector manufacturers provide thermoelectrically cooled packages and controllers specifically designed for their detectors.

5.2 Ultraviolet-Visible Region (UV-VIS)

5.2.1 Photomultipliers. Photomultipliers have been most extensively used for their high sensitivity, particularly in the UV region. Types with different cathode materials are selected, depending on the spectral region of interest and the sensitivity desired. The detector's active surface should be uniformly irradiated so that its response will be fairly independent of slight movements of the irradiating

beam and so that fatigue can be minimized. To avoid linearity errors and damage to the tube, the anode voltage should be set such that the maximum anode current is less than 10^{-6} A. To improve the signal-to-noise ratio, some types of photomultipliers can be cooled to -10 to -20 °C by using devices such as thermoelectric coolers. The stability of the high voltage power supply for photomultipliers is very important because the gain of a photomultiplier tube typically varies as the seventh power of the supply voltage. For best results, the voltage stability shall be less than 0.01% per hour. In addition, for consistent measurements, the high voltage should be applied to the tube before actual use for a stabilization period specified by the manufacturer.

5.2.2 Silicon Photodiodes. Silicon photodiodes are now being used commonly in spectroradiometers, and are recommended for use in the region of 250 nm to 1,100 nm, although the sensitivity of silicon photodiodes below 350 nm, coupled with normally low signals in the UV, normally makes their use impractical for measurements in the UV-A and UV-B regions. High quality silicon photodiodes, combined with a carefully designed current-to-voltage converter, have a sufficient sensitivity for common spectroradiometry (visible region). Most silicon photodiodes have much better linearity and stability than photomultipliers.

An amplifier for a silicon photodiode shall use a current-to-voltage converter (transimpedance amplifier) that provides input impedance close to zero. To achieve lowest noise level, it is recommended that the amplifier be built into the detector housing, and that an operational amplifier with low input-bias current and low offset-voltage drift be used. Temperature control may be incorporated to enhance detector and amplifier stability.

5.2.3 Other Photodiodes. There are other photodiode materials that can be used in the UV range. Some of these materials are silicon carbide (SiC), gallium nitride (GaN), and aluminum gallium nitride (AlGaN). Each has a spectral response range that covers from 200 nm up to 360 nm, depending on the construction. In addition to these materials, avalanche photodiodes can be used, which are high-speed, high sensitivity photodiodes using an internal gain mechanism that functions by applying a reverse voltage.

5.3 Near-Infrared (NIR) Region

5.3.1 Germanium Photodiodes. Germanium photodiodes are used in spectroradiometers covering the region of 800 nm to 1,800 nm. High quality germanium photodiodes combined with a carefully designed current-to-voltage converter have a sufficient sensitivity for common spectroradiometry. An amplifier for a germanium photodiode shall use a current-to-voltage converter that provides input impedance close to zero. To achieve lowest noise level, it is recommended that an operational amplifier with low input-bias current and low offset-voltage drift be used. Temperature control may be incorporated to enhance detector and amplifier stability.

5.3.2 Indium Gallium Arsenide (InGaAs) Photodiodes. Indium gallium arsenide (InGaAs) photodiodes are used in spectroradiometers covering the range of 800 nm to 1,700 nm. InGaAs is becoming more popular than germanium as a detector, mainly due to lower dark current. InGaAs also has a lower multiplication noise when used in avalanche photodiodes. InGaAs sensitivity can be extended to 2,600 nm. Similar amplification and temperature control as for germanium detectors may be required.

5.3.3 Other NIR Photodiodes. For the near infrared region longer than 1,100 nm, photovoltaic detectors, indium arsenide (InAs), indium antimonide (InSb), or photoconductive detectors such as lead sulfide (PbS) photodiodes can be used. Many detector manufacturers provide pre-amplifiers and detection schemes, which may include chopping the light to optimize the performance of their detectors.

6.0 Correction Methods

6.1 Flicker and Corrections

The light output of all light sources operating on AC, including incandescent lamps, is modulated in time (ripple), normally at twice the AC drive frequency. The magnitude of the modulation is small for incandescent (including tungsten-halogen) lamps; it is usually significantly larger for discharge lamps. For solid-state

lighting devices, the amount of flicker can be small or very large (100% modulation) depending on the driver design. This ripple may not be completely removed by the detector amplifier circuit if the time constant of the circuit is not long enough, or if the integrating time is not a multiple of the period of the frequency. In this case, the detector output signal shall be sampled by an integration over a period that is an integer multiple of the period of the power supply frequency. For array spectrometry, it is important that the measurements are integrated over several periods of the frequency. In order to satisfy this condition, the use of a neutral density filter may be required to increase the integration time of the array spectrometer.

6.2 Bandpass Correction

The bandpass of a spectrometer can cause significant errors in color measurement if the bandpass is larger than 5 nm. Even with a 5-nm bandpass, the uncertainty due to the bandpass can be the dominant term for special samples and for applications that require low uncertainties. One method of correction is presented in ASTM E308-18.⁹ This method can be applied only for object color measurement and only for a data interval of 10 nm or 20 nm, and it requires that the bandpass of the instrument is triangular and its width equal to the data interval. The weighting factor tables (Tables 6.1 – 6.36) of ASTM E308-18 are used to reduce bandpass errors effectively. The corrected tristimulus values are obtained by multiplying the spectral weighting factors (selected for the desired reference illuminant) by the reflectance factor data of the sample.

A second method is the “Stearns and Stearns method.”¹⁰ This method was developed as an analytical solution of the relationship between the true spectral values and the neighboring points of measured values, with the spectral distribution within the instrument bandpass modeled as quadratic with wavelength. The correction gives values at zero bandpass. (Table 6 of ASTM E308 is based on this method.) This method requires that the bandpass be a symmetric triangular function and its bandpass matched with the scanning interval wavelength step. The corrected spectral value S_i is simply calculated from the neighboring five points of the measured values M_i as:

$$S_i = (M_{i-2} - 12M_{i-1} + 120M_i - 12M_{i+1} + M_{i+2})/98$$

A third method is the extended method for bandpass correction, published by Ohno¹¹ and Gardner.¹² In real instruments, the instrument bandpass is not a triangular shape, and its width does not match the scanning interval. It is often seen that the bandpass of a spectroradiometer, designed to be constant, can vary as much as 20%. At 20% deviation, the reduction of the error with these methods will be about half or less. In some cases, the bandpass function might resemble a Gaussian function, a trapezoid, or an asymmetric triangle, which would also add deviation from the requirement. In such conditions where the bandpass and scanning interval are not well matched, ASTM E308¹⁰ or the Stearns and Stearns method¹¹ cannot be used. This improved method is now available to be applied to any arbitrary bandpass function. As this is a numerical approach, the solution can be given for any shape of bandpass, which may be nonlinear, asymmetric, and/or not matched with the scanning interval, with the assumption that the measured spectrum is a smooth function (quadratic within the bandpass). This method also provides accurate wavelength calibration for an asymmetric bandpass (often found in array instruments).

6.3 Stray Light and Stray-Light Correction

Stray light in spectrometers is the measured quantity of light that reaches the detector from a source other than that intended or selected wavelength. Therefore, stray light can cause erroneously high measurement results. Stray light is more of a problem with spectrometers than with monochromators due to size constraints. In general, stray light is evenly distributed along the entire length of the array; however, because the spectral distribution of the most common calibration source (the tungsten-halogen lamp) and the spectral responsivity of the array peak at about 900 nm, the stray light appears to be concentrated in the UV. The minimum requirement for stray light shall be at least three orders of magnitude smaller than the peak. Stray light at least four orders of magnitude smaller than the peak typically reduces the uncertainty contribution from stray light to a negligible level.

For many array spectrometers where the maximum wavelength measured is near 780 nm or 830 nm, the potential for stray-light errors is high because the

spectrometer is calibrated by an incandescent source, which has a large near-infrared contribution. If a source is then measured that does not have a similar spectrum, the blue signals will have large stray-light errors. The best solution is to have a cutoff filter in the system that eliminates the near-infrared light for the calibration source and any source to be measured.

For array systems that have no moving parts and have a fixed input geometry (e.g., optical fiber), a correction methodology has been developed by Zong et al.¹³ The basis is that the ratio of the flux of a particular wavelength of light detected by the desired detection element to that detected by a non-desired detection element is constant. By using narrow band lasers that have a bandpass smaller than the pixel wavelength resolution, a spectrum line spread function can be collected for many wavelengths. From the spectrum line spread function, the stray-light distribution function is calculated. The stray-light distribution functions can be assembled into a stray-light correction matrix. This matrix then can be applied to the collected spectrum to remove the contribution of stray light. (Refer to Zong et al.¹³ for a more detailed discussion.)

7.0 Calculations

7.1 Spectral Radiometric Quantities

When a standard source and a test source are measured for radiometric quantities such as spectral irradiance, spectral radiance, or spectral total radiant flux, the spectral quantities of a test source, represented by $Q_t(\lambda)$, are obtained from:

$$Q_t(\lambda) = R_t(\lambda) \cdot Q_s(\lambda)/R_s(\lambda),$$

where:

$R_t(\lambda)$ is the detector output reading for the test source

$Q_s(\lambda)$ is the spectral radiometric quantity of the standard source

$R_s(\lambda)$ is the detector output for the standard source

$Q_s(\lambda)/R_s(\lambda)$ refers to the spectral calibration factor of the spectroradiometer

$Q_t(\lambda)$ is obtained in the absolute unit used in the $Q_s(\lambda)$ data

For point sources, the spectral radiant intensity $I_s(\lambda)$ can be obtained from the spectral irradiance $E_s(\lambda)$ from:

$$I_s(\lambda) = E_s(\lambda) \cdot d^2,$$

where d is the distance (meters) between the geometric center of the source and the point where the spectral irradiance is measured.

7.2 Radiometric Quantities

Radiometric quantities such as irradiance, radiance, and total radiant flux can be obtained from the data of spectral radiance, spectral irradiance, and spectral total radiant flux, respectively, if these quantities are obtained in absolute units. The radiometric quantity of the test source is obtained by integrating over the wavelength range of interest.

7.3 Photometric Quantities

Photometric quantities such as luminance, illuminance, and total luminous flux can be obtained from the data of spectral radiance, spectral irradiance, and spectral total radiant flux, respectively, if these quantities are obtained in absolute units. The photometric quantity of the test source, represented by Q_v , is obtained from:

$$Q_v = K_m \int_{360}^{830} Q_t(\lambda) V(\lambda) d\lambda,$$

where:

$V(\lambda)$ is the spectral luminous efficiency

K_m is the maximum spectral luminous efficacy, 683 lumens/watt

It is common, however, to integrate over the region 380 to 780 nm, since the $V(\lambda)$ function at the far ends of the visible spectrum is so low. For example, the photometric quantity of an incandescent lamp in the region 780 nm to 830 nm is less than 0.001%. Therefore, unless the light source under test has a very large portion of its energy in this region, the error in leaving out this data will be negligible.

8.0 Reporting Requirements

The test report shall list all significant data. The report shall also list all pertinent data concerning conditions of testing and type of test equipment. Typical items reported are:

- Measurement date
- Identification and operating condition of test source
- Spectral quantity measured
- Type of spectroradiometer or monochromator, input optics, and detector used, including manufacturer and model number
- Standard source identification
- Spectral region, bandpass, and wavelength interval
- Wavelength scanning method (continuous or discontinuous)
- Table of wavelengths and spectral quantities with absolute units (if data are absolute)

The following values (using SI units) shall also be reported when applicable:

- Chromaticity coordinates [(x, y) and/or (u', v')] and correlated color temperature
- Color rendering indices (e.g., CRI, R_a , R_{fr} , and/or R_g , as applicable)
- Photometric quantities

ADDITIONAL READING

International Commission on Illumination (CIE). CIE 84-1989, The Measurement of Luminous Flux. Vienna: CIE; 1989.

International Commission on Illumination (CIE). CIE S 017/E:2011, ILV: International Lighting Vocabulary. Vienna: CIE; 2011.

International Commission on Illumination (CIE). CIE Research Note: A reference action spectrum for ultraviolet induced erythema in human skin. CIE J. 1987 Jun;6(1):17.

International Standards Organization (ISO) and International Commission on Illumination (CIE). ISO 23539:2005(E)/CIE S 010/E:2004: Joint ISO/CIE Standard: Photometry – The CIE System of Physical Photometry. Vienna: CIE; 2004.

Koller LR. Ultraviolet Radiation, 2nd ed. New York: John Wiley & Sons, Inc.; 1965:187, 271-273.

Kostkowski HJ, Reliable Spectroradiometry, Ch. 3. New York: John Wiley & Sons, Inc.; 1998.

National Bureau of Standards. NBS Special Publication SP 250-1, Spectral Radiance Calibrations. Washington, DC: National Bureau of Standards; 1987.

National Bureau of Standards. NBS Special Publication SP 910-4, Self-Study Manual on Optical Radiation Measurements; Part 1, Concepts. Washington, DC: National Bureau of Standards; 1979.

National Institute of Standards and Technology. NIST Special Publication SP250-89, Spectral Irradiance Calibrations. Washington, DC: National Institute of Standards and Technology; 2011.

Phillips, R. Sources and Applications of Ultraviolet Radiation. New York: Academic Press, A Subsidiary of Harcourt Brace Jovanovich; 1983.

REFERENCES

- 1 Sliney D, Wolbarsht M. Safety with Lasers and Other Optical Sources. New York: Plenum Press; 1980.
- 2 Illuminating Engineering Society. ANSI/IES RP-27-20, Recommended Practice for Photobiological Safety for Lamps and Lamp Systems. New York: IES; 2020.
- 3 Jerome CW, Jerome, Eby J. Chromaticity computation. Illum Engineering. 1970 Jan;65(1).
- 4 Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Feb 4).
- 5 Illuminating Engineering Society. ANSI/IES LM-78-20, Approved Method: Total Luminous Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: IES; 2020.
- 6 Ohno Y. Measurement of bandpass for array spectroradiometers. In: Proc. 27th Session CIE; 2011 Jul 11-15.
- 7 International Commission on Illumination (CIE). CIE 063-1984, The Spectroradiometric Measurement of Light Sources. Vienna: CIE; 1984.
- 8 Ohno Y. Spectral color measurement. Ch. 5 in: Colorimetry: Understanding the CIE System. Schanda J (ed.). New York: John Wiley & Sons, Inc.; 2007.
- 9 ASTM International. ASTM E308-18, Standard Practice for Computing the Colors of Objects by Using the CIE System. West Conshohocken, Penn.: ASTM International; 2018.
- 10 Stearns EI, Stearns RE. An example of a method for correcting radiance data for bandpass error. Color Res Appl. 1988;13(4):257–259.
- 11 Ohno Y. A flexible bandpass correction method for spectrometers. In: Proc AIC 2005, Granada, Spain; 2005 May 9–13:697–700.
- 12 Gardner JL. Bandwidth correction for LED chromaticity. Colour Res Appl. 2006;31:374-380.
- 13 Zong Y, Brown SW, Johnson BC, Lykke KR, Ohno Y. Simple stray light correction method for array spectroradiometers. Appl Opt. 2006;45/6, 1111–19.
- 14 International Commission on Illumination (CIE). CIE 214:2014, Effect of Instrumental Bandpass Function and Measurement Interval on Spectral Quantities, CIE 214:2014.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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